

REVIEW

The potential of transcranial magnetic stimulation for population-based application: a region-based illustrated brief overview

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The awareness of the global trends in neuroscience study, especially in the cognitive neuroscience field, should be increased. One notable approach is the use of transcranial magnetic stimulation (TMS) not only as a research tool but also as a choice for treatment and rehabilitation in neurological disorders, such as post-stroke hemiplegia, visuospatial neglect syndrome, Alzheimer's disease (AD) and psychiatric conditions such as major depression and schizophrenia. All of these occur in significant numbers in highly populated regions. This paper briefly discusses the basic protocols and potential benefits of using TMS with the aim of providing insight that is useful in the design of future public health strategies in highly populated regions with a large neurocognitive burden of disease where this technique is currently underemployed.

KEYWORDS: cognitive neuroscience, transcranial magnetic stimulation, neurocognitive disease

Background and Introduction

Cognitive neuroscience, since its emergence as a field in the 1980s, has grown astonishingly worldwide, and has contributed greatly to understanding the interactions between brain and behavior in the normal population as well as offering important insights for treatment and rehabilitation of patients with cognitive deficits. We take Indonesia and the Southeast Asia region as an example of the public health burden of the sorts of disorders in which cognitive neuroscience may offer significant advances in diagnosis and treatment. This region, like many others, has substantial neurocognitive health problems. Indonesia is the world's fourth most populous country with over 246 million people, and the Southeast Asia region as a whole comprises over 600

million people. Within this large population, stroke represents a large burden of disease. The pattern of clinical pattern data taken from 28 hospitals in Indonesia over a 6-month period in 1996 showed that strokes occurred in 35.8% of elderly patients with a 19.9% recurrence rate [1]. A more recent study revealed that stroke in Indonesia is contributing to 15.4% of all deaths [2] and it is the leading cause of death not only in Indonesia but also in the Southeast Asia region as a whole [3]. Although this data does not include incidences of stroke that did not result in hospitalization, it clearly gives an outline of the high rate of strokes seen. It is also important to note the possibility that general estimates of stroke incidence in developing countries significantly underestimate actual levels because of limited access to health services, meaning that a particularly high proportion of morbidity and mortality of people affected by stroke is left unreported [4]. The burden of stroke in Indonesia is highlighted by a study in central Aceh, Indonesia, in which, in addition to other health considerations, stroke was also found to affect patients' cultural and religious duties [5].

In addition to the high incidence rate of stroke, there are several neurocognitive problems related to mental

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health which, in a developing country like Indonesia, are strongly related to poverty and poor physical health [6]. For example, the number of people suffering from depressive disorder in Indonesia is high. A screening study of an elderly population in three Asian countries found that the prevalence of depression, determined using a cut off point of 5/6 on the geriatric depression scale (GDS-15), was 33.8% in Indonesia, 17.2% in Vietnam and 30.3% in Japan [7].

Although, like in many countries, there are high levels of cognitive and psychiatric disorders, cognitive neuroscience research and its application still plays a minor part in the local research community. It is always desirable to, where possible, provide proper, timely and effective treatment. Therefore, here we aim to provide better insight with regard to the use of noninvasive brain stimulation techniques such as *transcranial magnetic stimulation* (TMS) for greater application in both the laboratory and in clinical settings. This paper briefly discussed the potential benefit of TMS as a simple, effective, non-invasive and cost-effective approach to be used in the community. We aim to highlight important information that may be useful in the design of future public health policies and strategies not only for the illustrated region, but also for other highly populated regions with a large neurocognitive burden of disease.

TMS brief overview

Here we briefly describe the development, safety considerations and basic stimulation techniques of TMS in order to provide an overview of the use of this technique. Until recently, the study of the effects of brain lesions in humans, an important approach in investigating the necessity of cortical regions in behavior, was only possible in patients and other individuals who had suffered brain damage (typically as a result of accidents). Investigating the cognitive consequences of brain lesions in patients is hampered by several limitations. No two lesions are the same with regard to their location, size and functional impact, and lesions are rarely small enough to be confined to a single functional area in the brain. In addition, as a result of cortical reorganization, the cognitive consequences of a lesion may not be stable over time and effects seen as a consequence may give a misleading impression of the function of the damaged area.

In recent years, several modalities have been used to study brain function, such as functional scans using *positron emission tomography* (PET) or *single photon emission computed tomography* (SPECT) and more recently *magnetic resonance imaging* (MRI). These can provide information on whether there is fluctuation of cerebral perfusion as a consequence of performing a task, allowing inference of regions of cortical activation [8, 9]. These techniques generally have good spatial resolution. However, discussion of modern technology in cognitive

neuroscience often centers on the ideal of having tools with both good spatial resolution and good temporal resolution.

TMS is a tool that can allow for evaluation of the role of a brain region in cognitive function, so providing an alternative to lesion-based investigations, and has good spatial characteristics and excellent resolution in the temporal domain [10]. A typical approach using TMS employs stimulation for noninvasive direct local interference with cortical processing in a specific brain area, thus temporarily creating an effect frequently described as a “virtual lesion” in conscious healthy volunteers (although it is worth noting that while effects can be analogous to lesion effects, it is essentially stimulation resulting in disruption of activity in the area over which it is delivered). Observation of resulting behavioral changes provides causal information about the functional relevance of a brain area in a specific task or process. Using this approach, some neuropsychological conditions can temporarily be mimicked in healthy volunteers under controlled experimental conditions [11]. According to Day, Dressler [12], the mechanism by which TMS affects the brain is through magnetic stimulation inducing current flow under the stimulating coil in the horizontal plane and therefore most effectively stimulating horizontally oriented neurons, such as *interneurones*, pyramidal tract axon collaterals and afferent axons from cortical and subcortical sites. Although the ability of TMS to penetrate cortex is not more than approximately 2 cm in depth as shown by Bohning et al. [13] in a combined TMS/functional-MRI study, this still leaves a wide range of potential investigations where areas covered by this resolution are relevant.

Importantly, there is currently no evidence of undesirable short-term or long-term effects related to the use of TMS, and hence it is commonly described as a “relatively painless method” of stimulating the brain non-invasively [14,15]. However, this does not mean safety should be ignored and there are some safety issues that should be addressed when using TMS. For instance, and primarily of relevance to data quality, TMS causes significant sensory sensations that can nonspecifically interfere with task performance. These include a loud clicking sound as the stimulator is discharged, as well as potential stimulation of cranial nerves and the direct activation of facial and neck musculature [15]. To minimize the effects of the loud clicking sound, both in terms of data collection and, importantly, in terms of minimizing potential health effects, the recommendation is to use approved hearing protection (earplugs or ear muffs).

More important is the issue of potential seizures as a consequence of brain stimulation. It is reported that the risk of seizure during one type of commonly used stimulation (theta burst stimulation, TBS, see further) is as low as 0.02% and other high-frequency (hf) rTMS protocols have resulted in seizures in less than

0.1% of patients [16]. Importantly, most of the cases in which seizures occurred were before the introduction of defined safety limits [17] and the number of incidences since the introduction of guidelines has been very low (there is also some indication that the use of pro-epileptogenic medications may be a factor to take into account in these). This suggests that, when following appropriate guidelines, TMS is associated with minimal risk.

Additional issues to be addressed are magnetic field exposures for subjects or patients as well as for operators. Currently there is no evidence of a risk of long-term exposure. Finally, the effects of forces and magnetization on other devices also need to be considered, due to the interaction of the effects of the TMS coil with ferromagnetic objects, such as cochlear implants, skull plates, or pacemakers [17]. Clearly these interactions should be avoided by not employing TMS in subjects fitted with such objects.

There are several stimulation protocols commonly used when delivering TMS. These include single-pulse, double-pulse and repetitive TMS (low and high frequency) and TBS. Single-pulse TMS can be used, for example, in mapping motor cortical outputs and can also be used to answer questions about the effects of factors such as empathy on the motor system by measurement of TMS motor evoked potentials (MEP) [18]. Paired-pulse stimulation can be delivered to either a single cortical target or to two different brain regions using two different coils. This allows it to be used to both investigate the motor system and as well as study cortico-cortical interactions. For example, a study by Pascual-Leone and Walsh [19] started investigations of the roles of interactions between visual areas V5 and V1 in perception and awareness of visual motion.

“High-frequency” rTMS typically refers to stimulus rates of more than 1 Hz, and “low-frequency” (lf) rTMS to rates of 1 Hz (or less) [17,20]. The additional repetitive stimulation technique involves bursts of high frequency (50 Hz) stimulation delivered at a rate of 5 Hz and is termed TBS. This may have different effects depending on the exact combination of parameters [21]. In the intermittent TBS pattern (iTBS), a 2 s train of TBS is repeated every 10 s; in the intermediate paradigm (imTBS), a 5 s train of TBS is repeated every 15 s; the continuous paradigm (cTBS) consists of a 40 s train of uninterrupted TBS.

In general, lf (1 Hz) rTMS and cTBS will produce lasting inhibitory effects, whereas iTBS induces facilitatory effects [17,22,23]. This means that rTMS may be used to either enhance certain cognitive processes or to down regulate activity in specific brain regions depending on the parameters employed. For example, high frequencies have been typically reported to result in an increase of MEP size [24] and a study has demonstrated

a facilitatory effect specific for action word naming following hf-rTMS (20 Hz) of the left prefrontal cortex [25]. This latter effect was clarified in a patient study in which the 20 Hz hf-rTMS protocol was able to improve the linguistic skill in a primary progressive aphasia left hemisphere damaged patient [26]. The neurophysiological mechanisms responsible for rTMS-induced facilitation remain unclear, although they most likely relate to the activation of impeded pathways or inhibition of maladaptive responses [27]. However, in a detailed account of the effects of rTMS trains on the human motor cortex, Pascual-Leone, Valls-Solé [28] showed that the inhibitory effects observed at 10 Hz rTMS are likely due to post-activation refraction or direct activation of inhibitory intracortical interneurons. Suppression of excitability that persists after cTBS and the facilitatory effect after iTBS are found to rely on *N*-methyl-D-aspartate receptors (NMDARs) to produce after-effects in, for example, the motor cortex, resulting in induction of Ca^{2+} influx which triggers a series of reactions leading to long-term changes in synaptic strength [29].

Applications of TMS

Since its development as a viable experimental tool, investigators have used TMS for numerous functions, including (but by no means limited to): investigating parietal neglect, studies of the perception of visual motion, perceptual priming, the role of synchronized cortical discharge, visual conscious awareness, visual search and study of interactions between cortical areas [9,14,30,31]. TMS can also be combined with other techniques such as electroencephalography (EEG) and fMRI. Combination with EEG provides the possibility to non-invasively probe the brain's excitability, time-resolved connectivity and instantaneous state [32]. Although TMS induces a strong electrical field that can saturate recording amplifiers, there are several strategy to minimize the disruption this causes using a TMS-compatible EEG system which can include gain-control and a sample-and-hold circuit that locks the EEG signal for several milliseconds immediately post-TMS; using slew-rate limited preamplifiers, allowing continuous data recording and resulting in a short-lasting artifact; and using TMS-compatible DC amplifiers which allows continuous data recording without signal saturation [33,34].

Combination with fMRI-diffusion imaging, offers a way to investigate changes in the functioning of distinct parietal-frontal pathways, amongst others, after stroke and during recovery, promising a stimulating future for research with direct clinical relevance [35]. In the future, multimodal approaches using region-of-interest based functional mapping, combined with neural and behavioral individual analysis techniques could also be

employed when carrying out TMS experiments with the aim of identifying numerous frontoparietal topographic areas within each subject [36].

An example of a research area benefitting from the use of TMS is the topic of visuospatial neglect. A “neglect-like” effect can be produced by temporary disruption using TMS which can reverse the typical slight leftward bias seen in healthy subjects into a rightward bias. This effect mimics what can be seen in a right hemisphere stroke patient. Several TMS studies have used experimentally induced spatial deficits in healthy subjects by applying TMS over right posterior parietal cortex (rPPC) [30,37] to, for example, explore the spatial role of PPC using hf-rTMS [11,38]. These studies emphasized investigation of the neglect mechanisms and the typical right hemisphere dominance associated with them, but a study by Szczepanski and Kastner [36] has provided evidence that using single pulse TMS over left PPC can also shifts behavioral spatial bias (leftward in this case), which is more consistent with an interhemispheric competition theory. They also demonstrated the “neglect-like” effect of TMS over rPPC, producing a rightward bias when subjects perform the landmark task. In addition, Bien et al. [39] found left hemifield extinction following 20-Hz triple-pulse rTMS over rPPC and this was severely modulated when they displayed competing stimulus in the right hemifield. Other brain areas, such as right temporo-parietal junction (TPJ), have also been shown to be involved in neglect and for this area single pulse TMS over induces an extinction-like behavioral pattern for the contralateral hemifield [40].

TMS applications in patient studies

Here we briefly discuss potential benefits of TMS in terms of use in neuro-rehabilitative protocols in both neurological and psychiatric patients. It has been shown that there are significant roles to be played by TMS in clinical settings. TMS can potentially be used to therapeutic effect in patients where it seems to promote exogenous plasticity that, combined with endogenous mechanisms, may prove to have a “potentiation effect” on the effects of cognitive rehabilitation [41]. For instance, in Alzheimer’s disease (AD), patients’ performance on a naming task shows facilitation with TMS. High-frequency (20 Hz) rTMS was found to be beneficial both in mild and severe AD patients, possibly due to the presence of a compensatory mechanism following stimulation based on the recruitment of right hemispheric resources to support the residual naming performance [42].

A method for determining stimulation parameters (both in experimental and therapeutic approaches) is to use of motor cortex excitability threshold as the

baseline of TMS intensity given to subject brain areas and there are some studies that have used motor excitability in models of cognitive decline in neurological and psychiatric diseases. The resting motor threshold (rMT) is an indicator of motor cortex excitability in which the amplitude variation of the test MEP allows the evaluation of inhibitory and facilitatory intracortical phenomena [43]. In AD patients, the rMT was found to be significantly reduced as compared to healthy elderly people [22], an effect that is also observed in vascular dementia [44]. Another motor cortex derived parameter is the short latency afferent inhibition (SAI), an inhibition seen in MEPs 20–25 ms after a cortical TMS test pulse as a consequence of an electric conditioning pulse applied on the median nerve at the wrist. This can be used as a putative marker of central cholinergic activity [45] and is found to be significantly smaller in patients than in controls, providing neurophysiological evidence of central cholinergic dysfunction early in the course of AD [46].

In depressive disorders, a study has also been carried out to compare cortical excitability before and after hf-rTMS treatment. It was found that treatment resulted in the reduction of interhemispheric differences in cortical excitability in the treatment group when compared to control group and a clinical benefit was demonstrated in terms of an improvement in neuropsychological test scores [47]. A recent study has also showed that the bilateral rTMS treatment over the dorsolateral prefrontal cortex (DLPFC) is effective in treating the sleep alterations seen in drug-resistant depression. Specifically, hf (10 Hz) rTMS treatment over left DLPFC and low frequency (1 Hz) over right DLPFC were effective in inducing anti-depressant effects as well as resulting in an alpha power decrease in the REM sleep over the left DLPFC [48]. It has also been reported that lf-rTMS has acute effects and results in maintenance of efficacy in drug-resistant bipolar depression [49]. A study has shown that patients with severe depression appear to be cognitively better 1 week after a course of hf (10 Hz) rTMS than after 1 week of a course of unilateral electroconvulsive therapy (ECT) [50]. Consistently, right lateral prefrontal hf-rTMS has been reported to reduce depression for at least 8 h following stimulation [41]. TMS has also been shown to be effective and safe with few side effects in the treatment of major depression by delivering TMS stimulation over the left DLPFC [51,52].

TMS has also been evaluated in as a treatment in schizophrenia. A meta-analysis study revealed that lf-rTMS to the left temporo-parietal cortex (TPC) results in robust therapeutic effects on auditory hallucination symptoms, but small effects were also found on positive symptoms [53]. However, the use of a deep hf-rTMS technique using an H-coil over left DLPFC showed

a significant decrease of the schizophrenia assessment score after the period of treatment [54].

In post-stroke patients study, a study by Schwerin et al. [55] revealed that the application of high-intensity dual pulse TMS on stroke patients with unilateral hemiparesis drastically increased the occurrence rate of contralesional and ipsilesional MEPs in the paretic limb and produced strong facilitation across muscles. The application of 1 Hz rTMS to the primary motor cortex of the unaffected hemisphere resulted in a significant faster simple and choice reaction times of their affected hand [56]. A study of acute hemiplegia patients found that 3 and 10 Hz rTMS (as an add-on intervention to normal physical and drug therapies) was able to improve the stroke severity and functional ability scores as compared with sham rTMS [57]. Using the same parameters, Khedr et al. [58] found that rTMS was able to improve hand grip and shoulder abduction by 54% and 58%, respectively, followed an increase of functional scores. More importantly, this study revealed that the improvements were sustained 1 year after treatment.

The presence of visuospatial neglect as a symptom, where patients may not perceive part of the visual field, is an important predictor of poor functional recovery in stroke patients and it may greatly restrict patients' daily activities [59]. Specifically, a patient with visuospatial deficits experiences failure to acknowledge or explore stimuli in the contralesional side of space [60,61]. It occurs in approximately 43% of patients after right hemispheric stroke [62]. Using the TBS protocol, Koch et al. [63] found that TMS can be a potentially effective approach, presumably able to induce long-lasting changes in the excitability of the left hemisphere parieto-frontal circuits to balance with the damaged right hemisphere in visuospatial neglect patients. Furthermore, Nyffeler et al. [64] found that two TBS trains significantly increased the number of perceived left visual targets for up to 8 h following stimulation, and 4 TBS trains increased performance for up to 32 h.

For a condition like multiple sclerosis (MS), a study has found that TMS was useful in explaining the mechanism of neuronal alteration in MS by making use of single and paired pulse TMS to elicit rMT in the contralateral hand target muscles. Specifically, paired-pulse TMS was employed to probe intracortical inhibition (ICI). They found that relapsing MS patients showed increased thresholds and reduced silent period (SP) durations, while displaying a lack of ICI and the remitting MS patients showed prolonged SPs with normal rMTs [65].

TMS cost-effectiveness potential

The research and application of TMS in the laboratory as well as in clinical settings (as a neuro-rehabilitation

technique) must be considered to a greater extent, not least for potential budget savings it may offer and certainly for its potential to provide large benefits to patients. This is particularly the case in developing countries. The economic impact of treatment and rehabilitation for the patient as well as the quality of treatment and rehabilitation has to be taken into account. There is considerable evidence from developed countries that both inpatient and outpatient rehabilitation services have a significant effect on improving patient functioning [66]. It has been argued that TMS has cost-effective potential when it comes to its use in neuro-rehabilitation compared to the outcomes and costs of pharmacotherapy treatment [67]. In comparison with ECT for depression treatment, a single session cost of rTMS was lower than the cost of a session of ECT, although it is important to note that over all treatment sessions there were no treatment cost differences [68,69]. However, rTMS has advantages for maintenance treatment in terms of safety and tolerability for patients [70]. For central pain syndrome treatment, when compared with epidural motor cortex stimulation (MCS), although the efficacy of MCS is slightly better than TMS as shown by the mean reduction in pain as indexed by a visual analog scale (VAS) (3.34 vs. 2.14), the first treatment mean costs for MCS are 20 times more expensive than TMS [71]. There is a need for further large-scale, adequately randomized trials of rTMS comparing the cost-effectiveness with other techniques as well as with pharmacotherapy methods in treating particular cognitive problems.

Recently, the innovation in TMS has reached an important level with newly developed portable and battery-powered magnetic stimulators which are relatively inexpensive [72], although it has some drawbacks in terms of higher pulses rate, such as loss of power and issues related to heating. In the future, such innovations may prove suitable for the purposes of population-based TMS application.

Conclusion

To conclude, this paper briefly illustrates the potential of TMS application both in research and clinical settings as an excellent non-invasive targeted brain stimulation method. TMS has been used widely in 'virtual lesion' studies to mimic brain damage conditions and also been shown to be a favorable treatment modality for neurological deficits such as post-stroke hemiplegia, visuospatial neglect syndrome, Alzheimer's disease and some psychiatric conditions such as major depression and schizophrenia. Considering these substantial benefits, consideration should be given to TMS as an alternative modality of treatment to be made more widely

available clinically. Hence, broader embrace of TMS by the research and clinical communities, particularly in regions where it is relatively unused may yield advantages in the treatment and rehabilitation of cognitive impairments. More research in this field is needed to provide a truly evidence-based service to patients.

Declaration of Interest

The authors report no conflict of interest. The authors alone are responsible for the content and writing of this paper.

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